

4DM70 Analysis and design of networked dynamical systems

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Group Number: 5

INTRODUCTION

This report presents the results and analysis of the 4DM70 Group Assignment, which focuses on two problems in the subject of networked dynamical systems: network synchronization and multi-robot coordination. In the first part, synchronization behavior in networks of Hindmarsh-Rose (HR) model neurons is investigated, both theoretically and through experimental validation using an experimental setup. The assignment involves exponential convergence and strict semipassivity of the HR model, followed by analysis of practical synchronization conditions under varying coupling strengths and time delays. Additionally, partial and cluster synchronization patterns in larger networks are investigated.

In the second part of the assignment, a multi-robot coordination problem based on Voronoi coverage control is presented. The goal is to develop a control strategy that enables a group of unicycle robots to form an optimal coverage configuration while avoiding collisions. This involves designing a safe control algorithm, analyzing its stability, and validating the approach through MATLAB simulations and physical experiments using TurtleBots.

PROBLEM 1.1 HR MODEL PROPERTIES

$$\begin{aligned}\dot{y}(t) &= 100(-y_i^3(t) + 3y_i(t) - 8 \\ &\quad + 5z_{i,1}(t) - z_{i,2}(t) + E + u_i(t)) \\ \dot{z}_{i,1}(t) &= 100(-y_i^2(t) - 2y_i(t) - z_{i,1}(t)) \\ \dot{z}_{i,2}(t) &= 0.5(4y_i(t) + 4.472 - z_{i,2}(t))\end{aligned}\quad (1)$$

A. Using the Demidovich condition, show that the internal dynamics (1) are exponentially convergent

A system is exponentially convergent if there exist symmetric positive-definite $(n - m) \times (n - m)$ matrices P and R , and a constant $\delta > 0$ such that

$$P \left(\frac{\delta q}{\delta z}(z, w) \right) + \left(\frac{\delta q}{\delta z}(z, w) \right)^T P \leq -R \leq -\delta I$$

for all $(z, w) \in \mathbb{R}^{n-m} \times \mathcal{W}$. Here q can be defined as:

$$q = \dot{z} = \begin{bmatrix} \dot{z}_{i,1} \\ \dot{z}_{i,2} \end{bmatrix} = \begin{bmatrix} 100(-y_i^2(t) - 2y_i(t) - z_{i,1}(t)) \\ 0.5(4y_i(t) + 4.472 - z_{i,2}(t)) \end{bmatrix}$$

Such that:

$$\frac{\delta q}{\delta z} = \begin{bmatrix} -100 & 0 \\ 0 & -0.5 \end{bmatrix}$$

As P is symmetric positive-definite, P can be defined as $P = P^T = I$, which gives:

$$\begin{aligned}P &\begin{bmatrix} -100 & 0 \\ 0 & -1/2 \end{bmatrix} + \begin{bmatrix} -100 & 0 \\ 0 & -1/2 \end{bmatrix} P \\ &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -100 & 0 \\ 0 & -0.5 \end{bmatrix} + \begin{bmatrix} -100 & 0 \\ 0 & -0.5 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} -200 & 0 \\ 0 & -1 \end{bmatrix}\end{aligned}$$

This results in:

$$R = \begin{bmatrix} 200 & 0 \\ 0 & 1 \end{bmatrix}$$

a symmetric positive-definite matrix with

$$\delta = \min\{200, 1\} = 1 > 0$$

showing that the internal dynamics of the HR neuron model are exponentially convergent.

B. Use the radially unbounded and positive definite storage function to show that the HR model neuron is strictly semipassive from u_i to y_i

The storage function is defined as:

$$\begin{aligned}S &= \frac{1}{200}y^2 + \frac{1}{80}z_1^2 + \frac{1}{4}z_2^2 \\ \dot{S} &= \frac{1}{100}y\dot{y} + \frac{1}{40}z_1\dot{z}_1 + \frac{1}{2}z_2\dot{z}_2\end{aligned}$$

Where the derivative is defined as:

$$\begin{aligned}\dot{S} &= -y^4 + 3y^2 - 6y + 5z_1y - z_2y + Ey + uy \\ &\quad - \frac{5}{2}z_1y^2 - 5yz_1 - \frac{5}{2}z_1^2 + yz_2 + 1.118z_2 - \frac{1}{4}z_2^2\end{aligned}$$

where $E = 3.3$. Thus, the derivative becomes:

$$\dot{S} = uy - y^4 + 3y^2 - 4.7y - \frac{5}{2}z_1y^2 - \frac{5}{2}z_1^2 - \frac{1}{4}z_2^2 + 1.118z_2$$

Since $\dot{S} = uy - H$, where $H > 0$ for strict semi-passivity. Within the system, H can be defined as:

$$H = y^4 - 3y^2 + 4.7y + \frac{5}{2}z_1y^2 + \frac{5}{2}z_1^2 + \frac{1}{4}z_2^2 - 1.118z_2$$

Since $y^4 \pm \frac{5}{2}y^2z_1 + \frac{5}{2}z_1^2 > \frac{1}{4}(y^4 + z_1^2)$, H can be rewritten as:

$$H > \frac{1}{4}(y^4 + z_1^2) - 3y^2 + 4.7y + \frac{1}{4}z_2^2 - 1.118z_2 = \tilde{H}$$

Here, $\tilde{H}$ can be rewritten as

$$\tilde{H} = \frac{1}{4}(y^4 + z_1^2) - 3y^2 + 4.7y + \frac{1}{4}(z_2 - 2.236)^2 - 1.2499$$

Since z_1 and z_2 are positive squared terms, these terms are positive definite. To get the y-axis lower and upper bounds, the following can be computed for $\tilde{H}$:

$$\tilde{H} > \frac{1}{4}y^4 - 3y^2 + 4.7y - 1.2499$$

where

$$\frac{1}{4}y^4 - 3y^2 + 4.7y - 1.2466 > 0$$

when $y < -4.1077$ or $y > 0.3383$.

Moreover, to get the bounds of z_1 and z_2 , the inequality

$$\frac{1}{4}z_1^2 + \frac{1}{4}(z_2 - 2.236)^2 > -\frac{1}{4}y^4 + 3y^2 - 4.7y + 1.2499$$

can be evaluated for the maximum value of the right-hand side of the inequality on the domain $-4.1077 \leq y \leq 0.3383$. This can be computed using the derivative of the right-hand side:

$$-y^3 + 6y - 4.7 = 0$$

Resulting in

$$y = -2.7739, y = 0.9082, \text{ and } y = 1.8657$$

where only $y = -2.7739$ is within the domain. Now evaluating the inequality:

$$\begin{aligned} \frac{1}{4}z_1^2 + \frac{1}{4}(z_2 - 2.236)^2 &> 22.5694 \\ z_1^2 + (z_2 - 2.236)^2 &> 90.2775 = 9.501^2 \end{aligned}$$

This shows the z_1 and z_2 axes are bound by a circle with a center at $(z_1, z_2) = (0, 2.236)$ and a radius of 9.501.

Thus, it can be concluded that $H > \tilde{H} > 0$ outside the cylinder

$$z_1^2 + (z_2 - 2.236)^2 = 9.501^2 \quad \forall y \in [-4.108, 0.338]$$

PROBLEM 1.2: THEORY AND EXPERIMENT

Within this problem, the synchronization of eight electric HR neurons is investigated, where the neurons interact via coupling of the form:

$$u_i(t) = \sigma \sum_{j \in \mathcal{N}_i} w_{ij} [y_j(t - \tau) - y_i(t - \tau)]$$

Where σ is the coupling strength, τ is the time delay, and w_{ij} are the interaction weights.

Due to imperfections in the electronic components, a range of 0.25 is taken in between, which is defined as practical synchronization, which is considered as synchronization during the rest of the problem.

A. For each $\tau = 0, 1, \dots, 5$, determine a lower-bound and upper-bound for the coupling strength such that two coupled electronic HR model neurons practically synchronize.

Here, the synchronization for two neurons is given by the area of S^* in blue in Figure 1. Within this area, synchronization of the two neurons occurs. Since the $\sigma = 10$ is the upper-bound, the synchronization at $\tau = 0$ is maximized at $\sigma = 10$.

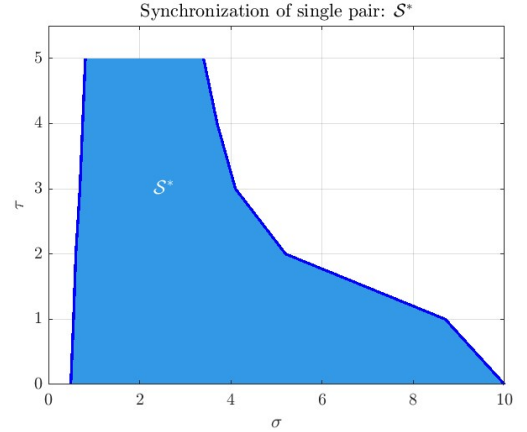


Fig. 1: synchronization area of 2 neurons

B. Predict the values of σ and τ for which the network of electronic HR model neurons below is practically synchronized.

The network, which is being discussed in B and C, can be seen in Figure 2. To predict the values of σ and τ for which the network synchronizes, the synchronization area can be scaled according to the following definition:

$$S_j^* := \left\{ (\sigma, \tau) \in \mathbb{R}_+ \times \overline{\mathbb{R}}_+ \mid \left(\frac{\lambda_j}{2} \sigma, \tau \right) \in \mathcal{S} \right\}$$

Where λ_j is the first non-zero eigenvalue of L . Where

$$L = \begin{bmatrix} 3 & -1 & -1 & 0 & -1 \\ -1 & 3 & -1 & 0 & -1 \\ -1 & -1 & 3 & -1 & 0 \\ 0 & 0 & -1 & 2 & -1 \\ -1 & -1 & 0 & -1 & 3 \end{bmatrix}$$

Which has the following eigenvalues: $\lambda_1 = 0$. Using λ_2 and λ_5 , the synchronization areas are computed for the smallest and largest non-zero eigenvalue. The region where both regions overlap is the region where full synchronization of the whole network occurs, which is indicated in blue. This can be seen in Figure 3.

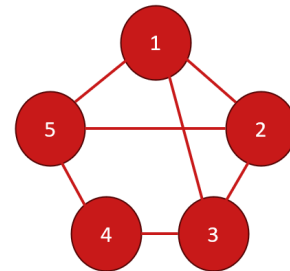


Fig. 2: Network, with undirected edges of weight 1

In order to see if this matches the actual synchronization area, measurements are conducted. The results of the measurements can be seen in Figure 4. Since the measured

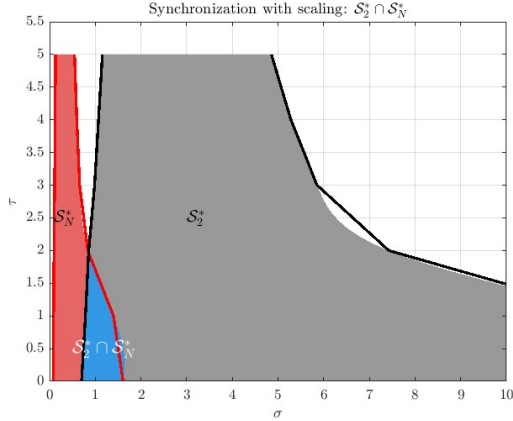


Fig. 3: synchronization area of scaled network

network of two neurons is scaled, while being limited by an upper-bound of $\sigma = 10$ for no delay, the theoretically calculated region is smaller than the measured region. Additionally, the scaling definition is a conservative calculation. Thus, it is fairly reasonable that the measured region is larger than the calculated region.

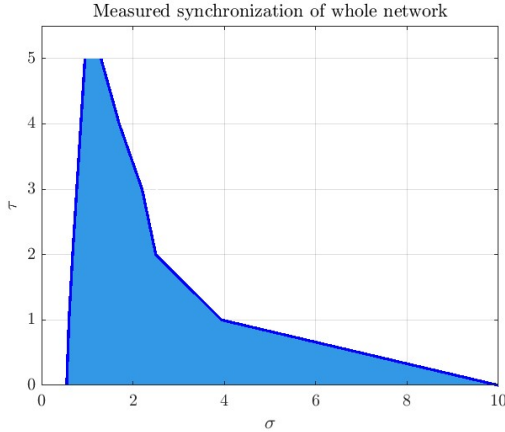


Fig. 4: Measured synchronization of whole network

C. Add one edge of weight 1 to improve synchronization. Explain how you determined the edge to be added.

Firstly, intuition is used to determine where it would be suitable to add an edge. Either between neurons 1 and 4, which would yield the same results due to symmetry, or between 3 and 5, an added edge would improve synchronization most. To understand which added edge would improve synchronization most, the λ_2 values of the new systems were investigated, as λ_2 is the algebraic connectivity. The higher the value of λ_2 , the larger the synchronization. Thus, when choosing which edge to add, the largest value of λ_2 is looked at. Since for the system with edge 1-4 the $\lambda_2 = 3$ and for the system with edge 3-5 the $\lambda_2 = 2$. Therefore, the added edge between neurons 1 and 4 yields the best synchronization results.

To validate whether this added edge helps to improve the

synchronization of the system, the synchronization bounds at $\tau = 5$ were measured. Before the added edge, the lower bound was $\sigma = 0.95$, which has now decreased to $\sigma = 0.70$. The upper bound has increased from $\sigma = 1.30$ to $\sigma = 1.40$. Thus, the synchronization has indeed improved by adding an edge between neurons 1 and 4.

D. Describe this mode of partial practical synchronization and give a corresponding permutation matrix

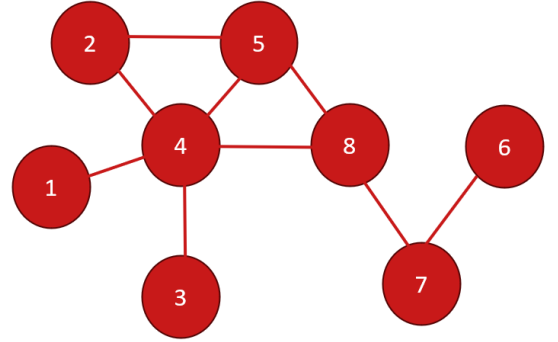


Fig. 5: New network, with undirected edges of weight 1

The new network of eight neurons, which is being discussed in D, E, F, and G, can be seen in Figure 5. Within this network, there occurs partial synchronization when $\sigma = 1.5$, while complete synchronization only occurs when $\sigma \geq 2.75$. The partial synchronization occurs between the neurons, which can be switched according to the permutation matrix Π , due to symmetry. The partial synchronization occurs between neurons 1 and 3, as can be seen in Figure 6.

$$\Pi = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

E. Specify the value of the time delay at the transition from practical synchronization to partial practical synchronization, describe the observed mode of partial practical synchronization, and give the corresponding permutation matrix.

The given network with coupling strength $\sigma = 3.5$ loses practical synchronization at $\tau = 0.6$ while partial practical synchronization is maintained between neurons 1 and 3. The permutation matrix is given by:

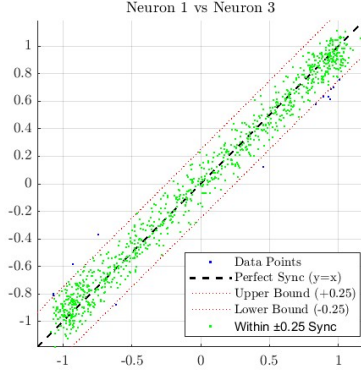


Fig. 6: synchronization between neuron 1 and 3

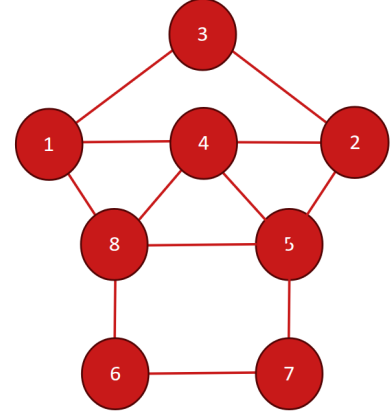


Fig. 7: Network with added edges 1-8 and 5-6.

$$\Pi = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

F. Explain your results of problems D and E, using the theory of partial synchronization.

To see whether partial synchronization occurs, the smallest eigenvalue of L with corresponding eigenvector in the range $(I - \Pi)$ is determined. Since $\lambda' = \lambda_3$, partial synchronization can be expected for appropriate values of σ , as $\lambda' \geq \lambda_2$.

G. Add at most two edges of weight 1 to the network such that, for some value of the coupling strength lower than required for full practical synchronization, you get practical partial synchronization of the desired form.

First, network symmetries that support the desired synchronization pattern where neurons 1 and 2, neurons 6 and 7, and neurons 5 and 8 synchronize, and neurons 3 and 4. Edge additions that would introduce new symmetries were examined in the existing network structure. The objective was to create a symmetry group that would include permutations (1,2), (6,7), and (5,8). This required introducing edges that created the needed symmetry, without accidentally inducing synchrony in 3 and 4.

After analyzing the symmetry structure, adding an edge between neurons 1 and 8, and neurons 5 and 6 created the desired synchrony, Figure 7, and partial synchronization pattern with the permutation matrix:

$$\Pi = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix}$$

This completes the objective to create a symmetry group including permutations (1,2), (6,7), and (5,8). To validate this design, the network dynamics with the modified topology were simulated. With the simulation, the desired practical partial synchronization form was confirmed to occur at a coupling strength of $\sigma = 0.68$.

PROBLEM 2

A. Design control (i.e., linear and angular speed) inputs for all three unicycle robots to perform safe Voronoi coverage control.

The control for the robots is split into attractive and repulsive potentials. Together, they give a potential function used to determine the new control variables. The control variables are converted to the linear and angular velocity for the robot.

The attractive potential is based on the distance between the robot and the goals and is scaled by a control gain κ .

$$V_i = \frac{1}{2} \kappa_i \|x_i - x_{goal}\|^2$$

The repulsive potential between two robots is based on the distance between the two robots, the radius of the robots r , and is scaled by a weight w_{ij} .

$$V_{ij} = \frac{1}{2} w_{ij} \|x_i - x_j - 2r\|^2$$

Combining the attractive and repulsive potentials, the following gradient H is constructed to create a safe control algorithm with a soft threshold scale k .

$$\frac{\delta H_i}{\delta x_i} = -\frac{\delta V_i}{\delta x_i} + \frac{1}{1 + k(V_{ij} + V_{ik})} \frac{\delta}{\delta x_i} (V_{ij} + V_{ik})$$

From the gradient of H , the linear and angular velocities of the robot can be computed with a linear gain k_v and the angular gain k_ω using

$$\begin{aligned} v &= k_v(v_x \cos(\theta) + v_y \sin(\theta)) \\ \omega &= k_\omega \text{atan2}(-v_x \sin(\theta) + v_y \cos(\theta), \\ &\quad v_x \cos(\theta) + v_y \sin(\theta)) \end{aligned}$$

To ensure the linear and angular velocities calculated and sent to the robot don't require higher wheel speeds than possible, the values are scaled according to the maximum linear and angular velocity of the robot using:

$$\begin{aligned} v &= \max\{\min\{v, v_{\max}\}, -v_{\max}\} \\ \omega &= \max\{\min\{\omega, \omega_{\max}\}, -\omega_{\max}\} \\ s &= \frac{|v|}{v_{\max}} + \frac{|\omega|}{\omega_{\max}} \\ \text{If } s > 1 : \quad &\begin{cases} v = \frac{v}{s} \\ \omega = \frac{\omega}{s} \end{cases} \end{aligned}$$

The determined velocities are finally sent to the robot via the ROS2 publisher. If the position of the robot is within some threshold value of the calculated Voronoi centroids, commands for zero velocity are sent.

B. Show that your control design ensures collision avoidance and is asymptotically stable

Collision avoidance is tackled within the repulsive potential V_{ij} . The potential takes the actual distances between the robots, being the center points minus the radii of both robots, into account when following a safe path. The path is followed based on $\frac{\delta H_i}{\delta x_i}$, where the attractive gradient to the goal is dominant when the robots are not close together. However, when the robots get in close proximity of each other, the potential grows, becoming dominant, leading the robots to move away from each other, as can be seen in Figure 8. Additionally, a saturation term is added in $\frac{\delta H_i}{\delta x_i}$, to prevent a too large influence of close proximity of two robots on each other and on the other robot.

To ensure stability, the coordination potential is asymptotically minimized by:

$$\dot{V} \leq 0$$

Where $V = H_i$, where $H_i = \frac{1}{2}\kappa_i\|x_i - x_{\text{goal}}\|^2 + \frac{\frac{1}{2}w_{ij}\|x_i - x_j - 2r\|^2}{1 + (\frac{1}{2}w_{ij}\|x_i - x_j - 2r\|^2)}$. and thus $\dot{V}$ can be written as:

$$\dot{V} = \frac{\delta V_i}{\delta x_i} + \frac{\delta}{\delta x_i} \frac{(V_{ij} + V_{ik})}{1 + k(V_{ij} + V_{ik})}$$

The partial derivatives are given by:

$$\frac{\partial V_i}{\partial x_i} = 2 \cdot \frac{1}{2}\kappa_i\|x_i - x_{\text{goal}}\| = \kappa_i(x_i - x_{\text{goal}})$$

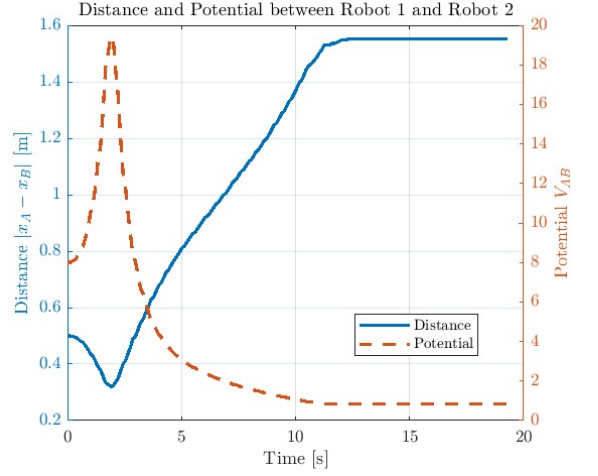


Fig. 8: Potential and distance between two robots over time

$$\frac{\partial V_{ij}}{\partial x_i} = w_{ij}(x_i - x_j - 2r), \quad \frac{\partial V_{ik}}{\partial x_i} = w_{ik}(x_i - x_k - 2r)$$

The total repulsive potential sum can be defined as:

$$V_r = V_{ij} + V_{ik}$$

The time derivative of the total potential is computed using the chain rule:

$$\dot{V} = \frac{dH_i}{dt} = \frac{\partial V_i}{\partial x_i} \cdot \dot{x}_i + \frac{d}{dt} \left(\frac{V_r}{1 + kV_r} \right)$$

Applying the quotient rule to the second term yields:

$$\frac{d}{dt} \left(\frac{V_r}{1 + kV_r} \right) = \frac{(1 + kV_r)\dot{V}_r - kV_r\dot{V}_r}{(1 + kV_r)^2} = \frac{\dot{V}_r}{(1 + kV_r)^2}$$

Where the derivative of V_r with respect to time is defined as:

$$\dot{V}_r = \frac{d}{dt}(V_{ij} + V_{ik}) = \frac{\partial V_{ij}}{\partial x_i} \cdot \dot{x}_i + \frac{\partial V_{ik}}{\partial x_i} \cdot \dot{x}_i = \frac{\partial V_r}{\partial x_i} \cdot \dot{x}_i$$

Therefore, the total time derivative becomes:

$$\dot{V} = \left(\frac{\partial V_i}{\partial x_i} + \frac{1}{(1 + kV_r)^2} \cdot \frac{\partial V_r}{\partial x_i} \right) \cdot \dot{x}_i$$

To ensure stability, the control law is used as:

$$\dot{x}_i = - \left(\frac{\partial V_i}{\partial x_i} + \frac{1}{(1 + kV_r)^2} \cdot \frac{\partial V_r}{\partial x_i} \right)$$

Substituting into $\dot{V}$ gives:

$$\dot{V} = - \left\| \frac{\partial V_i}{\partial x_i} + \frac{1}{(1 + kV_r)^2} \cdot \frac{\partial V_r}{\partial x_i} \right\|^2 \leq 0$$

Which always yields a negative result.

Since $\dot{V} \leq 0$ and $V \geq 0$, the potential H_i asymptotically decreases to a constant value where $\dot{V}_i$, $\dot{V}_{ij}$, and $\dot{V}_{ik}$ are zero. This guarantees convergence of robot x_i toward the goal while maintaining safe distances to the other robots x_j and x_k .

C. Demonstrate your multi-robot coordination algorithm in at least three different MATLAB simulations

Four distinct scenarios were created to evaluate the multi-robot coordination under varying conditions. Each scenario uses identical event locations, each spaced one meter apart: $p_1 = (1.58, 0)$, $p_2 = (0.73, 0.5)$, and $p_3 = (0.73, -0.5)$.

Scenario 1: The first scenario creates a baseline to evaluate the Voronoi coverage control of the robots. Figure 9 shows the dynamic updating of Voronoi centroids (denoted by $\times$) throughout running the control algorithm. Initially, a linear formation emerges in the group while moving toward their respective Voronoi centroids. Upon approaching the high event density region, the robots reconfigure into an equilateral triangle formation. This configuration represents the optimal coverage strategy for the specified event positions, where each robot governs a pie-shaped portion of the high-density area, with robot positions placed near the designated event points that define the distribution.

Scenario 2: This scenario examines collision avoidance under a challenging initial configuration where the robots are positioned in close proximity and facing each other. Figure 10 displays the Voronoi centroids corresponding to the initial robot positions.

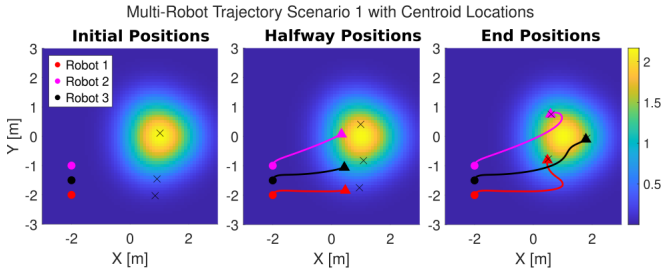


Fig. 9: Trajectory of the robots for scenario 1 showing the Voronoi coverage control and updating of the calculated centroids.

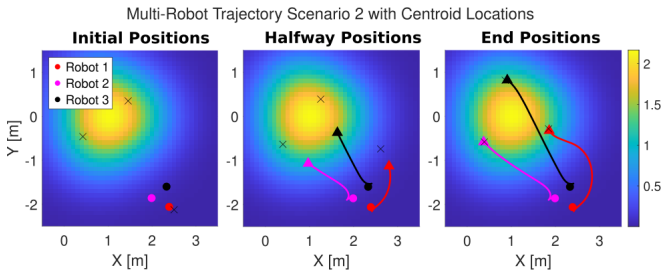


Fig. 10: Trajectory of the robots for scenario 2 showing the Voronoi coverage control and updating of the calculated centroids.

The triangular starting formation results in pie-shaped Voronoi cells for each robot. Consequently, the initial Voronoi centroid of robot 1 is located away from the high event density area with respect to itself. However, after Robots 2 and 3 move toward their respective centroids, the system converges to the same equilateral triangle formation observed in Scenario 1.

Scenario 3: This scenario evaluates collision avoidance when the robots start in a line formation pointing towards the event locations, where the first robot in the line has malfunctioned and is unable to move. Figure 11 demonstrates that Robots 1 and 2 successfully avoid the stationary Robot 3 while moving towards their designated Voronoi centroid, showing the collision avoidance working.

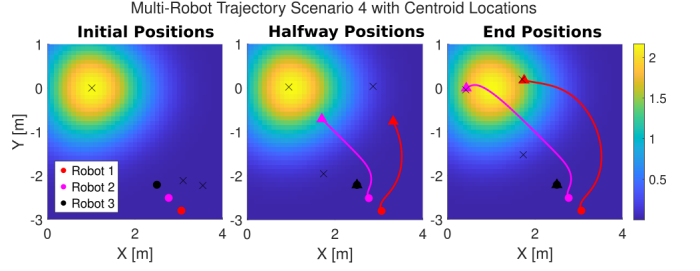


Fig. 11: Trajectory of the robots for scenario 3 showing the Voronoi coverage control and updating of the calculated centroids.

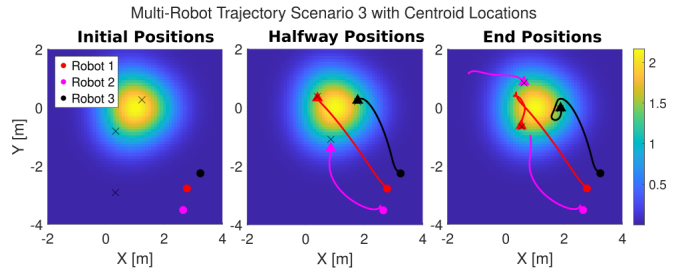


Fig. 12: Trajectory of the robots for scenario 4 showing the collision avoidance of the coverage control.

Scenario 4: The final scenario assesses the updating behavior of the Voronoi centroids and system robustness in unexpected situations. Initially, robots move from their starting positions to their optimal Voronoi coverage configuration. Subsequently, one robot is moved to the opposite side of the formation. Figure 12 shows that Robots 2 and 3 respond appropriately by adjusting their Voronoi coverage positions, demonstrating that the robot control creates a new optimal coverage configuration.

D. Test your multi-robot coordination algorithm in at least two different experiments using TurtleBots in the ME Robotics lab.

Similar to the experiments in the MATLAB environment, the save coverage control has also been tested in similar scenarios with the TurtleBots of the ME Robotics lab.

Scenario 1: This initial scenario validates the Voronoi coverage control on TurtleBot platforms and compares it against MATLAB simulation results. Figure 13 shows the updating Voronoi centroids (denoted by $\times$). The robots display similar patterns to the MATLAB simulation: initial linear formation followed by an equilateral triangle configuration.

Scenario 2: This scenario replicates the challenging initial configuration from the MATLAB study with the robots positioned in close proximity and facing each other. Figure 14

confirms successful collision avoidance on the TurtleBots and presents similar patterns as the MATLAB simulation experiment.

Scenario 3: The final scenario mirrors the MATLAB experiment in which one of the robots is moved unexpectedly. Figure 15 shows that Robots 2 and 3 appropriately adapt their Voronoi coverage positions, creating a new optimal formation that validates the system control in line with the results from the MATLAB experiment.

given event distribution, the final result remained consistent across the MATLAB simulation and physical TurtleBots.

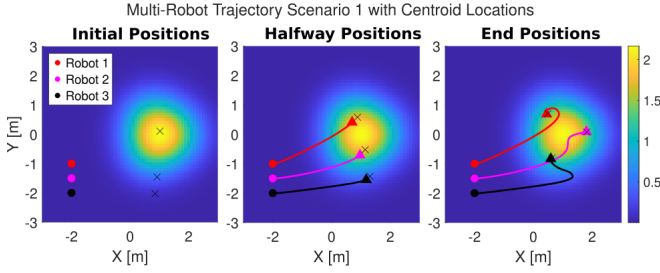


Fig. 13: Trajectory of the robots for scenario 1 showing the Voronoi coverage control and updating of the calculated centroids.

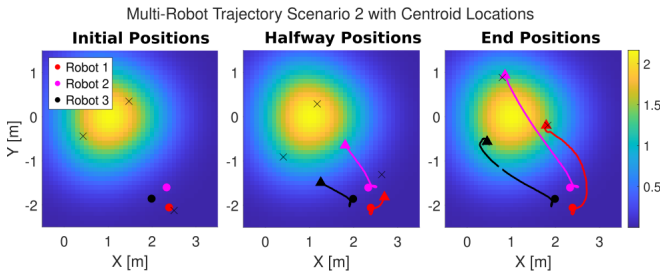


Fig. 14: Trajectory of the robots for scenario 2 showing the collision avoidance of the coverage control.

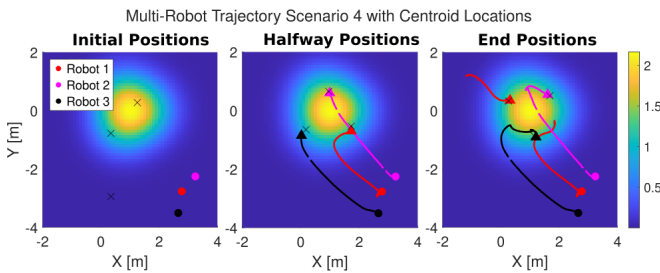


Fig. 15: Trajectory of the robots for scenario 3 showing the updating behavior of the Voronoi centroids and the robustness of the control.

Conclusion

With these experiments, the multi-robot coordination algorithm is shown to translate well from the MATLAB simulation to the physical TurtleBots, achieving similar coverage control configurations. While real-world dynamics and disturbances affected the speed and precision at which the robots were able to form the desired configuration for the